ASUS UGen AI League · Thunderstorm track (ASUS UGen Intel Arc Pro B70) · Workplace AI · Global Online final

Ledger Room

Checks a bookkeeping firm's vouchers before the VAT return. AI checks, code verifies, a person signs. One office box built from ASUS parts with a UGen B70; client data stays in the office.

Code: github.com/StuartYo/ledger-room

113 real invoices: the old ledger let 15 wrong rows through, Ledger Room let 0

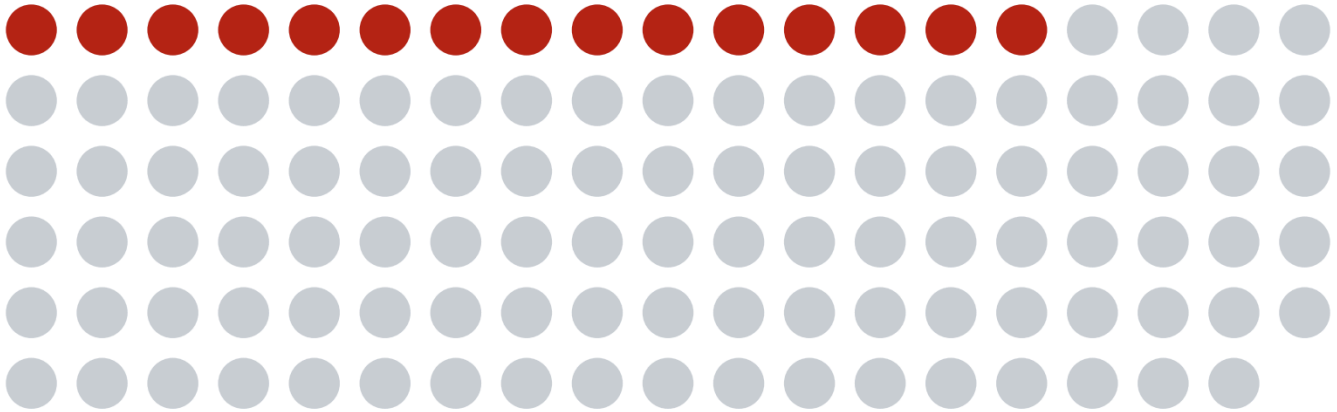
Team Ledger Room · SID 388470
Measured on the organizers' Arc Pro B70, the same GPU as the ASUS UGen B70.
Compared on the old ledger's three fields: invoice number, date, total.
Real-company numbers are aggregates, shown with the owner's consent. Demo screens use sample data.

THE PROBLEM

Keyed vouchers go wrong without anyone noticing. Caught before the tax office asks, a wrong return costs no fine

15 of 113

invoice rows were wrong in a real small company's ledger, typed with a general chat AI and checked once.



Wrong fields: 7 invoice numbers, 6 dates, 3 totals

NT$0 fine

when the firm corrects a return before any inquiry: tax plus interest only. Found by the tax office first: 0.5× the tax for a duplicate claim, 1× for a non-deductible one.

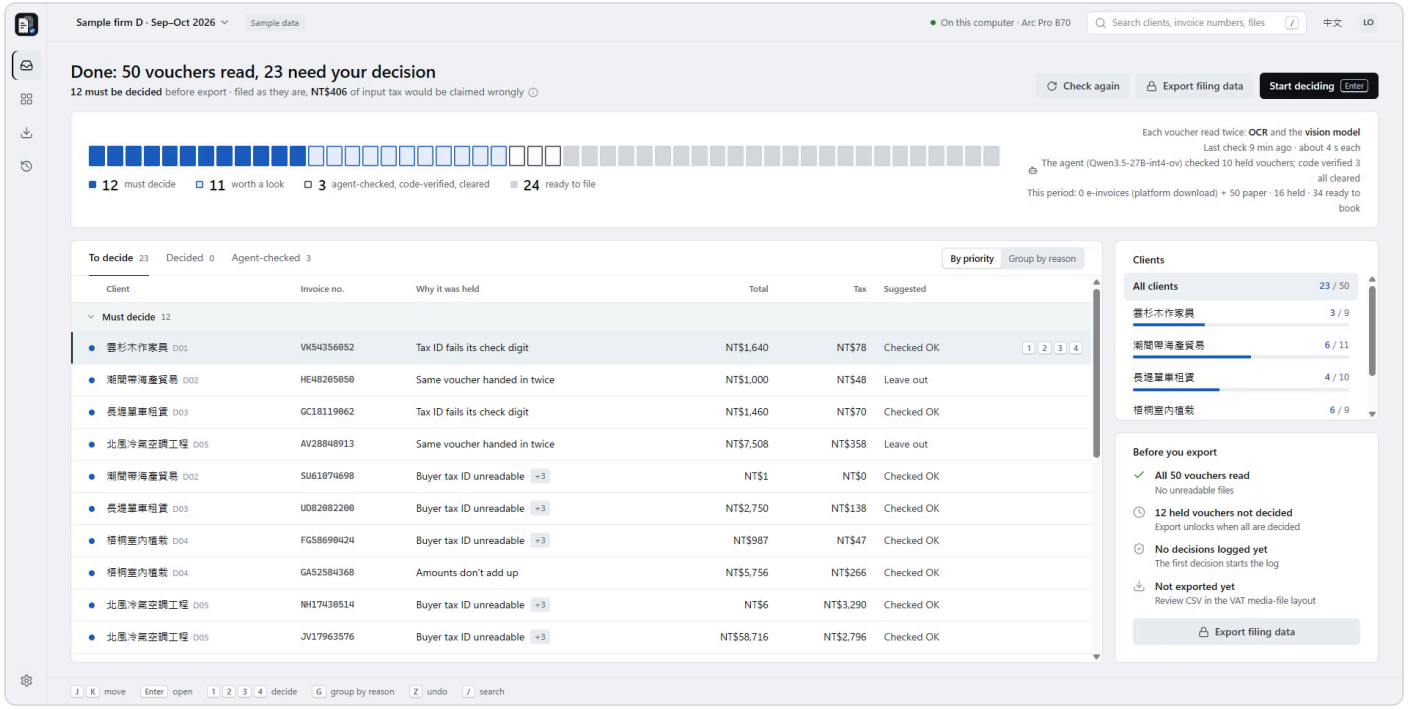
85,288

VAT violation cases in Taiwan in 2025; 25,600 were fined, NT$3.5 billion in total.

A real small company's petty-cash box, owner's consent, aggregates only. 113 of its 121 invoices pair with a ledger row; compared on the ledger's three fields. eval/results_real_company.json. Tax Collection Act Art. 48-1; MOF penalty reference table (VAT, 2024-11-21); MOF Fiscal Statistics Yearbook 2025, Table 3-29 (all VAT violations, including deliberate evasion). Sources: appendix A12.

THE PRODUCT

AI checks, code verifies, a person signs: the bookkeeper opens only what is still in doubt



Live run on the organizers' Arc Pro B70 host, synthetic sample firm: the 9B read all 50, the 27B agent re-checked 10 of the 16 held (one demo run; full experiment on slide 6).

The 9B reads every voucher
Staff press Check. OCR and the Qwen3.5-9B vision model read each scan.

Rules hold the doubtful ones
Two printed sources must agree. Six tax rules, each with its article.

The 27B agent re-checks
Only held ones: zooms in, re-reads, runs the sums, proposes a value.

Code verifies
A proposed value counts only with evidence from outside the model.

A person signs
The rest, each with its reason and law. Then the VAT filing file.

The 401 filing pair passes the MOF offline checking program (synthetic client). Detail: appendix A1, appendix A2. Sources: appendix A12.

WHAT THE BOOKKEEPER SEES

A person opens only the vouchers still held, and sees why each one was held



Same host run, synthetic voucher handed in twice: reason and law, evidence, one key to decide.

Why, and which law

the rule that held it and the article it rests on

The evidence

crops of number, total and date, and the earlier copy of the same voucher

One key to decide

Enter takes the marked suggestion

This one: handed in twice

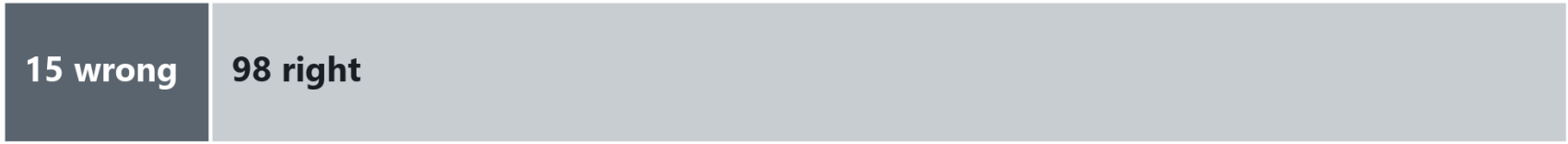
Claimed twice, the input tax is a duplicate claim: fined 0.5× the tax if the tax office finds it first.

Rule: VAT Act Art. 51(1)(5); MOF penalty reference table (duplicate input-tax claim, 0.5×). Synthetic client D02, invoice HE48205050 (also handed in for 2026-09). How the checks work: appendix A1.

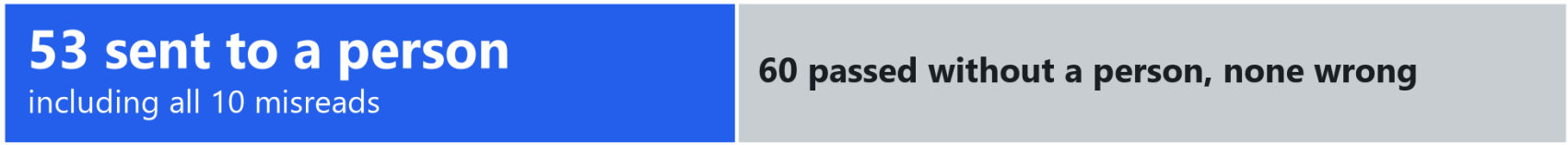
THE EVIDENCE

113 real invoices: the old ledger let 15 wrong rows through, Ledger Room let 0

The old ledger
chat AI, checked once
15 let through



Ledger Room
9B on the B70, strict reading
0 let through



58 photos: 0 through
public Taiwan photos, strict reading: 41, plus 17 new ones on a first run

MOF checker: PASS
the 401 filing pair passes the Ministry of Finance's own offline checking program, on a sample client

2,383 reads in 70 minutes
two B70 cards looping four test sets; every pass scored the same, 0 restarts

Compared on the old ledger's three fields: invoice number, date, total. Answer key frozen before any read. All six fields: appendix A3. Public photos: appendix A4. MOF offline VAT checking program 11500.01, forms 401 TXT + TET_U, 2026-11-08. 70 minutes: four synthetic test sets in a loop (eval/results_longrun_zoompad.json), appendix A6. Sources: appendix A12.

THE AGENT

The 27B agent cleared 19 of 52 held vouchers with strict reading, and none of the 19 was wrong

Today: 9B and rules

30 to a person

Code-only sum check

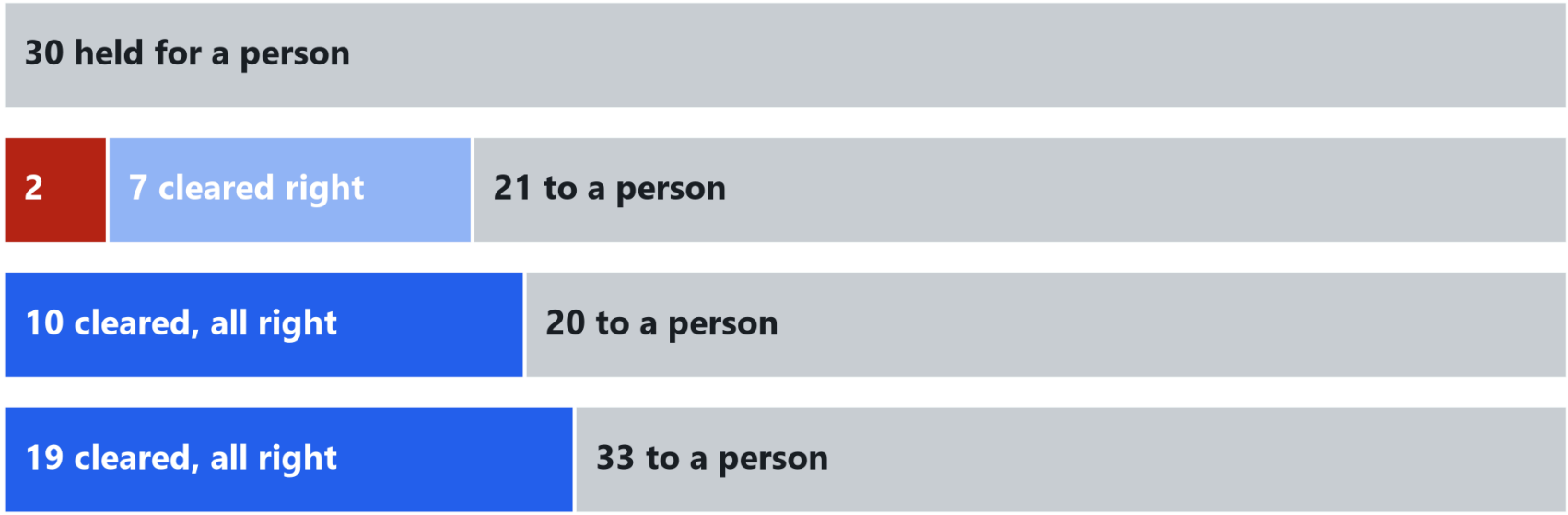
2 wrong cleared

27B agent, code verifies

0 wrong cleared

Same, strict reading: 52 held

0 wrong cleared



9 of 10
cleared because net + tax = total checked out in code; 1 on a zoomed re-read of the buyer's tax ID.

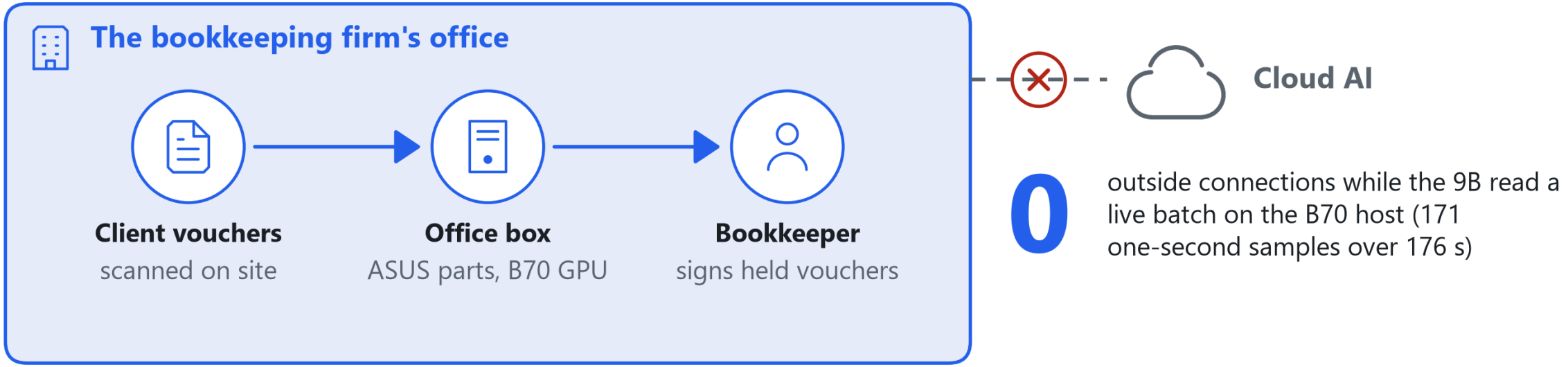
blind3: 12 → 6
held vouchers in six layouts never seen before, run once after the check rules were fixed.

8 s
a held voucher on average (18 s slowest), spent only on the held pile. Two runs, same result.

Synthetic test sets: 30 held of 204 vouchers in four sets; the organizers' host, second Arc Pro B70 (GPU.1), 2026-11-08 and 09, run twice with the same result. Rows 1–3: default reading (strict off), as the experiment ran (eval/results_agent_b70.json, "strict": false); row 4: strict reading, as real documents run (eval/results_agent_b70_strict.json); no cleared voucher was wrong: of the agent's 9 wrong proposals, code refused 8; the voucher carrying the other one (a date) stayed with a person. Detail: appendix A2.

WHY IT SITS IN THE OFFICE

Client records stay in the office: a cloud reader is outsourcing the firm must supervise



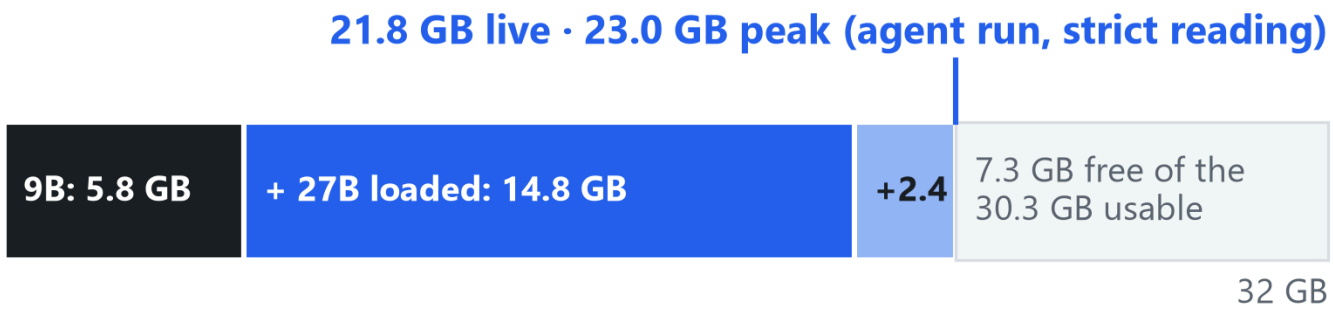
- A cloud reader, even one in Taiwan, is outsourcing the firm must supervise in writing (Art. 10).
- Sending data abroad adds Art. 11: tell each client where it goes. Google Cloud's Taiwan region lists no Gemini generation models.
- On a box the firm owns, nothing is outsourced and nothing leaves the office.

Regulations on the Security of Personal Data Files of Certified Public Bookkeepers and Bookkeeping and Tax Agents (law-out.mof.gov.tw GL011130), Arts. 10, 11. Vertex AI locations page, updated 2026-11-06. eval/results_offline_b70.json (code 3524145), measured on the organizers' Arc Pro B70 host. OpenVINO telemetry is switched off before OpenVINO loads, guarded by a test (tests/test_telemetry_off.py). Sources: appendix A12.

The 9B reader and the 27B agent stay loaded together on one Arc Pro B70: 23.0 GB of its 32 at peak

Card memory, both models loaded

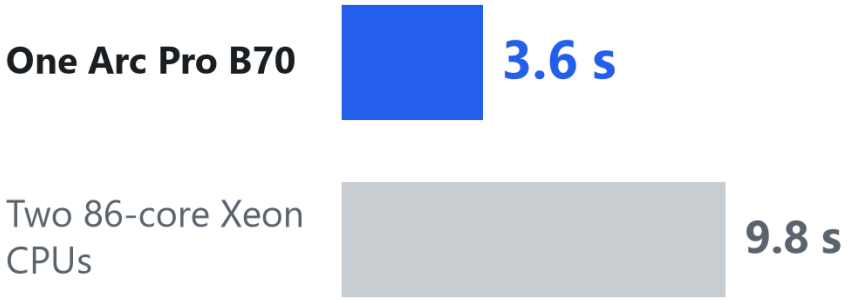
Both stay loaded for the whole period, so no model swaps between reading and checking.



Both loaded: 20.55 GB; working on the held vouchers: 23.0 GB at peak. Live: both models reading 20 documents. Driver reports 30.3 GB usable.

One card beats the host's two 86-core Xeon CPUs

Same code, same machine: seconds a voucher, 9B reading 12 held-out documents



The 27B agent adds 8 s, only on held vouchers.

Measured on the organizers' Arc Pro B70, the same GPU as the ASUS UGen B70: two cards: 2,042 vouchers an hour (1,031 + 1,011) for 70 minutes, no restarts; one reader keeps a card 91% busy · 194 W while reading · 187 Wh per 1,000 vouchers · p50 3.4 s · p95 4.1 s a voucher on one card (47 held-out documents). More: appendix A6.

Memory: live = one-card run, 9B then 27B reading 20 documents, 21.8 GB (results_one_card.json); peak = agent run on GPU.1, strict reading, 23.0 GB (results_agent_b70_strict.json, appendix A2); the driver also mapped 15.2 GB of system memory to the GPU. Seconds: OCR, 9B reading and zoomed re-reads together; CPU against B70 (GPU.1), 12 held-out documents, live re-run 2026-11-09, code 0ebd76a, 0 fields wrong on both. Organizers' host: Xeon 6787P, two Intel Arc Pro B70 (Intel's own boards), OpenVINO 2026.4.1. Sources: appendix A12.

WHO BUYS IT, AND WHY

Firms buy it to catch wrong claims before the tax office does: 8 catches a year pay for the box and the licence

85,288
VAT violation cases in Taiwan in 2025; 25,600 were fined, NT$3.5 billion in total (all kinds, deliberate evasion included).

8 catches
a year pay off the box and the licence in three years: 1 return in 38 for a 50-client firm. 2 a year pay the licence alone. Example: a NT$6,000 non-deductible input-tax claim found first by the tax office draws a NT$6,000 fine (MOF penalty table, 1×).

NT$20
per client a month, at least 50 clients (NT$12,000 a year). The box: NT$91,501–99,245, an ASUS UGen B70 card with an ASUS board, power supply and case.

- Fewer fines first: corrected before filing, no fine. MOF penalty table: 0.5× the tax for duplicate claims · VAT Act Art. 51: up to 5×.
- Less typing second: every voucher arrives already read; staff open only the held ones (53 of 113 on a real company's invoices).
- Paper and e-invoices alike: platform downloads run through the same rules; on a sample period all 18 planted errors were caught, 0 false holds.
- Market: about 8,600 bookkeeping and 2,435 CPA firms in Taiwan, first about 320 large ones. ASUS parts: NT$66,451–74,195 a box at retail.

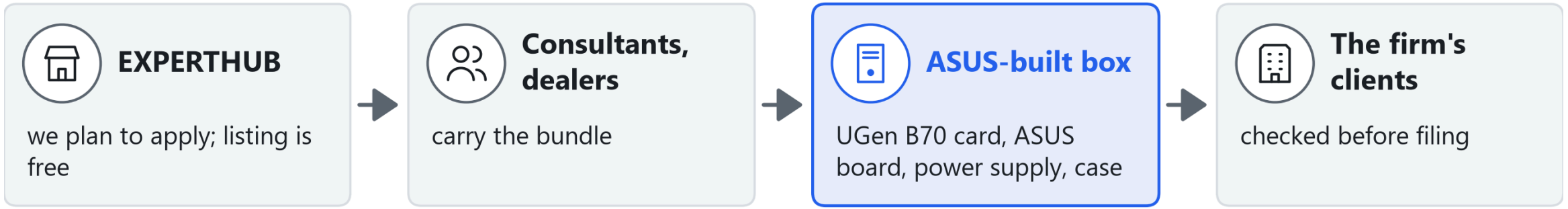
8 catches: box NT$91,501–99,245 + 3 × NT$12,000 licence = NT$127,501–135,245; 24 × NT$6,000 = NT$144,000; 50 clients × 6 returns a year = 300, 300 ÷ 8 ≈ 38. Example: MOF penalty reference table (VAT, 2024-11-21), Art. 19 case 1×; Kaohsiung tax office case (Economic Daily, 2026-06-11). Cases and fines: MOF Fiscal Statistics Yearbook 2025, Table 3-29. Held vouchers: eval/results_real_company.json, 9B, strict. Sample period: synthetic, 140 platform e-invoices and 44 paper (eval/results_period_check.json). More: appendix A7. Sources: appendix A12.

HOW IT REACHES FIRMS

How it reaches firms: an on-premises bundle on ASUS EXPERTHUB, built from parts ASUS already sells

0 of 4

None of the 4 finance apps on EXPERTHUB is built for firms that file VAT for many clients. EXPERTHUB targets 1.715 million small businesses in Taiwan.



Every box carries NT$66,451–74,195 of ASUS parts: NT$21–24 million for the first 320 firms, NT$0.73–0.82 billion across Taiwan's 11,035 bookkeeping and CPA firms.

- Armor EdgeAI, already on EXPERTHUB, is a general on-premises AI toolkit that runs only on ASUS GPU machines. Ledger Room would list the same way, for one job: a firm's VAT period, checked across all its clients.
- IFA, also listed, keeps one organisation's own books. Ledger Room serves the firm that files for many clients and keeps their records in its office.

experthub.asus.com, Finance & Expense, read in Chrome 2026-11-07; solutions 364 (Armor EdgeAI) and 145 (IFA), read 2026-11-08. EXPERTHUB target market and its bundle model: cnyes 2026-07-16; free listing: bnext 2026-07-17. Alternatives: appendix A8. Sources: appendix A12.

DESIGN DECISIONS AND NEXT STEPS

AI checks, code verifies, a person signs. Next: one bookkeeping firm, two filing periods

Doubt → agent, then a person

Money → code

First run → official

MEASURED ON THE ORGANIZERS' ARC PRO B70

0 of 113
wrong invoices let through, on the old ledger's three fields

19 of 52
held vouchers cleared by the agent, 0 wrong

23.0 GB
of 32 GB: the 9B and the 27B together on one card

3.6 s
a voucher; 9.8 s on the host's two Xeon CPUs

About 1,000
vouchers an hour a card; 70 minutes, 0 restarts

0
outside connections while reading

NEXT

Pilot: one firm, two filing periods
Counts the claims the firm corrects before filing (appendix A9).

Agent on real held vouchers
The firm's own held vouchers, with its earlier vouchers to look up.

Code and result files **github.com/StuartYo/ledger-room**

Same GPU as the ASUS UGen B70. Agent: synthetic test sets, strict reading (slide 6). Questions we expect: appendix A10 · Pilot plan: appendix A9 · Scope and next measurements: appendix A11 · Every number's source: appendix A12

APPENDIX A1 · HOW IT WORKS, IN DETAIL

Models read and investigate. Code checks every value before it counts

Read
OCR: PP-OCRv4
Qwen3.5-9B INT4
OpenVINO GenAI
on the Arc Pro B70
+ MOF platform file

→

Check in code
two printed sources must agree
6 tax rules
firm-wide duplicate index

→

Agent, held only
Qwen3.5-27B INT4
tools: zoom and re-read, sum check, UBN check digit, earlier vouchers, platform file

→

Code verifies
agent value counts only with evidence from outside the model

→

A person signs
the rest, with reason and law; then 401/403 filing file, voucher list, signed log

- The models fill in fields and propose values. Deductible, duplicate, over a limit: six rules decide, each citing its article (UBN check digit; same voucher twice; one voucher under two clients; VAT Act Art. 19 non-deductible; over the client's own limit; date outside the period).
- Strict reading, used for every real-data number in this deck: each field that goes into the filing needs two printed sources that agree (OCR, the 9B, the MOF list of issued invoice prefixes).
- The agent works only on held vouchers: at most six tool steps, temperature 0. For an amount, code wants net + tax = total with the other two amounts already confirmed or printed; for a number or a date, OCR on the zoomed crop must read the same digits. A number that merely appears on the page is not evidence: the model can take the net for the total.
- E-invoices come from the MOF platform's media filing download, without OCR.

The agent on 30 held synthetic vouchers: 10 cleared, 0 wrong, and the check rules we tuned on the way

Synthetic sets; 30 of 204 vouchers held by today's flow (9B)

Set	Docs	Held	Agent + code cleared	Wrong, cleared
Damaged prints (hard)	49	18	**4**	0
blind3, six unseen layouts	53	12	**6**	0
heldout2	55	0	–	–
heldout	47	0	–	–
All four sets	204	30	**10**	0

Check rule for agent amounts, tuned on the damaged set only

Version	Cleared	Wrong
v1: an amount counts if that number is printed on the page	5	1
v2: all three amounts printed on the page and adding up	2	0
v3, used: the three amounts add up; the other two are confirmed or printed	4	0

- Code-only shortcut on the same 30: 9 cleared, 2 wrong. In both, the 9B had misread net and tax; the agent read the right totals (3,224 and 2,531), the sums disagreed, and code held them.
- The agent proposed 35 values; 5 were wrong and code stopped all 5. 6 vouchers with only correct proposals stayed held for lack of outside evidence: v1 shows what a looser rule lets through. Earlier-voucher and platform-file tools: kept for real firms, where sellers repeat.

Run on the organizers' host, second Arc Pro B70 (GPU.1), 2026-11-08 and 09, run twice with the same result, default reading (strict off), as the experiment ran (eval/results_agent_b70.json); blind3 first ran after v3. Card memory peak: 22.4 GB here, 23.0 GB in the strict re-run. 8.0 s a held voucher on average, 18.0 s slowest, 2.2 tool steps. Strict re-run (eval/results_agent_b70_strict.json, code a975bd7): 52 held, 19 cleared, 0 wrong; code refused 8 of 9 wrong proposals; the other (date) stayed on a held voucher. Keys: appendix A12.

336 real vouchers by kind: what the 9B reads and what goes to a person

9B on one B70	Vouchers	Strict: to a person	With a wrong field	Strict: wrong, not held	Default: to a person / not held
E-invoice printouts	83	27	26	0	27 / 0
Triplicate uniform invoices	24	15	12	1	14 / 1
Duplicate register invoices	14	14	14	0	14 / 0
Utility and telecom bills	155	145	117	9	111 / 32
Postage receipts	34	34	11	0	34 / 0
Tax bills	14	13	5	1	11 / 1
Other receipts	12	9	8	0	9 / 0
All vouchers	336	257	193	11	220 / 34

- Reading, share of fields wrong: OCR alone 60%; 2B alone 68.1%, 9B alone 27.0%; Ledger Room with the 2B 36.1%, with the 9B 21.5%.
- Strict, whole box: the 2B sends 309 of 336 to a person and lets 7 through; the 9B sends 257 and lets 11 through, 9 of them utility bills; a bill is now booked from its e-invoice on the MOF platform, which leaves 2 through and 112 to a person (bill keywords written after seeing this box).
- 27B second reader on the 164 photos the 9B was unsure of: 21.5% to 18.2% of fields wrong, 241 to a person, still 11 through; no change on the public photos.

Real petty-cash vouchers of one small Taiwanese company, used with the owner's consent; aggregates only, no images, name or tax ID. Answer key: 4 labelers from the page images only, frozen before any read. Fields: invoice number, date, seller UBN, total, net, tax. 9B code fa8c027, 3.87 s a photo; 2B code b6fd8bd, 2.45 s. eval/results_real_company.json.

APPENDIX A4 · PUBLIC REAL PHOTOS

58 public Taiwan photos, strict reading: 0 let through

First run, 41 photos, 9B on one B70, code 99f36fc (3.33 s a photo)

	OCR + our rules	9B alone	Ledger Room
Key fields wrong	29 of 164	28 of 164	13 of 164
Photos with a wrong field	19	20	10
Photos sent to a person	38	0	5
Wrong or blank field, not sent to a person	1	20	5

Strict reading. 41: rescored after the fix. 17 new: first run

	41: 2B	41: 9B	17: 2B	17: 9B
Key fields wrong	18 of 164	11 of 164	19 of 66	15 of 66
Photos with a wrong field	13	9	10	8
Sent to a person	21	14	14	9
Wrong, not sent to a person	1	0	0	0

- The 5 on the first run: 3 had a blank invoice number or seller UBN, a code gap; 1 had a wrong seller UBN, flagged by the check-digit rule; 1 was a real misread, O where the invoice number prints Q.
- Since then a blank field goes to a person, and strict reading wants two agreeing printed sources for every filed field. On the same 41 photos: 0 let through, 14 sent to a person.
- 17 new Commons photos, answer key frozen before any run, first run with strict reading: 0 let through, 9 of 17 sent to a person.

Wikimedia Commons (CC BY-SA); keys labelled from the image only, frozen before runs; the 41 come from one uploader, mostly flat scans. Files: eval/results_real_tw_b70.json; eval/results_real_tw_strict.json (fa8c027), _2b.json; eval/results_real_tw2_strict.json (968f9e0), _2b.json.

APPENDIX A5 · BLIND TEST ON NEW LAYOUTS

Six layouts it had never seen: rule F1 0.84 on the first run, 1.00 after the reader fix

blind3: 53 synthetic documents in six new layouts, answer key frozen before any Ledger Room code ran on them. First official run, 9B on one B70, code 77fa568.

9B, same code, one B70	Held-out: 47 docs	blind3: 53 docs, new layouts
Rule micro F1	1.00	0.84: found 23, missed 4, 5 false alarms
Key fields wrong	0 of 319	40 of 326
Most often wrong	none	buyer UBN 10/26, net 10/26, tax 7/26, total 10/53
Sent to a person	0 of 47	11 of 53

After the reader fix (r9, 2aead27)	Held-out	blind3	Damaged set, first reader
Rule F1; key fields wrong; to a person	F1 1.00; 0/319 wrong; 0 of 47 to a person	F1 1.00; 18/326 wrong; 12 of 53 to a person	45/320 wrong; 18 of 49 to a person

- Invoice number split by OCR: code took the eight digits for the buyer's UBN. That caused all 5 false alarms and 3 of the 4 misses.
- Amounts written one digit per box: OCR read single digits and code let them override the model. All 9 such invoices still went to a person.

13
documents had a wrong key field

11 of 13
went to a person

2 of 13
were accepted: hotel statements with the seller UBN read as the client's own and the buyer UBN blank. Blank buyer UBN: input tax not claimed. Wrong seller UBN: a wrong value in the filing data.

Answer key eval/expected_blind3.json, frozen 2026-11-06. Raw file: eval/results_tiered.json; 13 / 11 / 2: eval/blind3_first_run_per_doc.json. Reader r9 rejoins split invoice numbers and sends a seller UBN equal to the client's own to a person. After-fix rows: eval/results_after_fix_r9.json and eval/results_after_fix_r9_blind3.json.

APPENDIX A6 · HARDWARE INTEGRATION

About 1,000 an hour per B70 for 70 minutes, no restarts

Organizers' host with two Intel Arc Pro B70 cards (Intel's own boards; the same GPU as ASUS UGen B70), OpenVINO 2026.4.1. Short runs: 47 held-out documents. 70 minutes: one 9B per card, four test sets in a loop. Targets frozen in the README before the runs.

Held-out set, 47 docs	One B70: Qwen3.5-9B INT4	One B70: Qwen3.5-27B INT4	Both B70s: one 9B per card
Read time per document	3.7 s (target ≤ 6 s)	5.2 s (target ≤ 12 s)	47 docs, 85 s for the batch
Latency p50 / p95	3.4 / 4.1 s (one reader)	–	3.5 / 4.5 s (2,383 docs, 70 min)
Documents per hour	1,034 with one reader 1,013, 993, 976 with 2, 3, 4 readers	692 with one reader	2,042 an hour (2,383 docs in 70 min; 1,031 + 1,011); before the fix 1,886
Key fields wrong; micro F1 (model alone)	0 of 319; F1 1.00 (0.20) same with 2 to 4 readers	0 of 319; F1 1.00 (0.47)	0 of 319, F1 1.00 70 min: every pass of every set scored the same (held-out F1 1.00; hard set F1 0 by design, nothing planted)
Card memory, peak	6.6 GB per reader (card counter)	17.0 GB; 15 GB on disk 9B + 27B together: 21.8 GB live · 23.0 GB peak (agent run, strict reading)	one 9B on each card
Power (xpu-smi, each second)	194 W while reading, 64 W idle; 187 Wh per 1,000 docs	–	197 + 191 W; 190 Wh per 1,000 docs (70 min)
Compute engine busy (DRM fdinfo)	–	–	91% / 91% of the time, 70 min (before the fix 86% / 84%)
GPU reader error (CL_OUT_OF_RESOURCES)	First: reload every 5 docs, 600–660/h Then: a fresh reader only after an error (blind3, reader r9: 883/h, 1 restart)	–	Traced to one image, the UBN zoom strip (852 × 140 px): 4 errors in 640 full reads, all on it; padded to 448 px, 0. 70 min after the fix: 0 restarts, 0 docs to a person. Before: 28 restarts, 3 to a person

eval/results_b70.json (3524145; 3.7 s mean = 973/h; 1,034, latency and power = eval/results_concurrency.json rows.0); eval/results_concurrency_2cards.json. Error and fix: eval/results_oor_repro.json. 70 min after the fix: eval/results_longrun_zoompad.json (LEDGER_ZOOM_MIN_H=448, c370a45; the default since a49cc19), eval/results_longrun_zoompad_util.json; before: eval/results_longrun.json, eval/results_longrun_util.json. Busy share from Xe DRM fdinfo (xpu-smi gave N/A); energy = sampled watts × wall clock. eval/results_tiered.json, eval/results_after_fix_r9_blind3.json.

APPENDIX A7 · THE NUMBERS BEHIND SLIDE 9

Fines, keying, the box and the market, in numbers

	Number	How we got it (our calculation unless a source is named)
Example: fines that pay the licence	2 a year	NT$20 × 50 clients × 12 = NT$12,000 ÷ a NT$6,000 fine (1×, non-deductible claim found first by the tax office). The pilot counts them in the firm
Example: fines that pay the box and the licence	8 a year (1 return in 38)	Box NT$91,501–99,245 + 3 × NT$12,000 licence = NT$127,501–135,245; 24 × NT$6,000 = NT$144,000 over 3 years. 50 clients × 6 returns a year = 300; 300 ÷ 8 ≈ 38
Paper invoices passing without a person	24% (9 of 38; 95% range 13%–39%)	One real company's triplicate and register invoices, 9B, strict. E-invoices come from the platform unkeyed, so they are left out
Payback on keying alone, 10,000 paper invoices a month	18.4–28.5 months	NT$0.65 saved an invoice (45 s × 24% × NT$219 an hour); box NT$91,501–99,245; licence NT$1,500–3,000 a month (75–150 clients at NT$20)
Break-even within 3 years	above 6,234–8,879 a month	At 30 s a voucher: 32–75 months; at 60 s: 13–18 (10,000 a month)
Postage, tax bills, receipts too	13% pass; 43–157 months	13 of 98 paper vouchers; 10,000 a month
Box cost a voucher over 3 years	NT$1.27–1.38	2,000 vouchers a month × 36; no power, upkeep or licence
Box without the card	NT$33,420	Core Ultra 5 225; ASUS B860M-E-CSM; 32 GB DDR5; 1 TB SSD; ASUS TUF 750 W (ASUS's pick for this card); ASUS Prime AP201 (coolpc.com.tw, 2026-11-08)
ASUS UGen B70 card	NT$58,081–65,825	Converted from €1,799 in Germany × 36.59; the low end swaps 19% German VAT for 5%
Two-card box (small-business tier)	NT$166,482–181,970	Two UGen B70 cards; ASUS PRO WS W880-ACE SE board, TUF 1200 W, Prime AP303
Market in Taiwan	8,600 + 2,435 firms	Bookkeeping firms (MOF, 2025) and CPA firms (FSC). First target about 320: 297 CPA firms above NT$10 million a year, 26 bookkeepers with 10,000+ records
ASUS parts for 320 / 11,035 firms	NT$21–24 million / NT$0.73–0.82 billion	NT$66,451–74,195 of ASUS parts a box (card, board, power supply, case) × 320; × 11,035 (8,600 + 2,435)

The pilot recomputes the pass share from the firm's own form 401 counts (appendix A9). Measured on Intel's own 230 W Arc Pro B70 board on the organizers' Xeon host, the same GPU as the 310 W ASUS UGen B70 in the box we price; we measure the ASUS-built box next. Held vouchers count as zero saving. Sources: appendix A12.

Others read invoices on premises; none we found checks a firm's clients against each other

Taiwan (vendor pages)	Duplicates across a firm's clients	Rules, each with its reason	Frozen public test	Who sees the documents	Where it runs
NewSoft GenAI-OCR: 22 voucher kinds; a firm is a customer	not stated publicly	not stated publicly	One firm: 'over 90%' right	not stated publicly	Cloud or on premises
3Q: for bookkeeping firms; platform e-invoices only, no AI	not stated publicly	Number gaps, deduction type	not stated publicly	The firm	PC or NAS in the firm
EzTax: for companies filing their own VAT; test version	Within one company	Art. 19, 84 expense types	'95%+' reading	not stated publicly	not stated publicly
AI Buuking: AI vouchers, NT$12–20 each; serves firms too	not stated publicly	Judgement calls left to staff	not stated publicly	The vendor's cloud	Cloud
Armor EdgeAI (on EXPERTHUB): general on-premises AI toolkit, six modules	not stated publicly	not stated publicly	not stated publicly	The business itself	ASUS GPU machines only
IFA (on EXPERTHUB): one organisation's books, OCR helper	not stated publicly	VAT and income tax modules	not stated publicly	not stated publicly	Cloud or on premises
Ledger Room	Yes: one index across all clients	Yes: six rules, each citing its article	Yes: keys frozen first; first runs published	The firm's own staff only	On premises, one box

- Ledger Room is the only one we found that checks duplicates across a firm's clients, with a frozen public test (searched 2026-11-07).

"not stated publicly" means we found nothing on the vendor's public pages, not that the feature is missing; vendor rows are the vendors' own claims. Read 2026-11-07: newsoft.com.tw/ocr-taiwan-invoice/, motoway.tw/products/acct-platform, eztax.com.tw, aibuuking.com; read 2026-11-08: experthub.asus.com/solution/detail/364 and /145.

Pilot plan: one bookkeeping firm, two filing periods

Setup

- One bookkeeping firm, two VAT filing periods (four months).
- The box sits in the firm's office. No client data leaves it, and nobody logs in to it from outside.
- Staff use Ledger Room beside their usual work. The firm's bookkeeper still signs and files every return.

Success measures

- Share of vouchers sent to a person, each period, before and after the agent.
- Wrong vouchers not held, counted against the return the firm finally files. Target: 0, with and without the agent.
- Hours spent keying and checking each period, recorded by the firm itself, before and during the pilot.
- Seconds a person needs to check one held voucher. The keying figures in appendix A7 count held vouchers as zero saving.
- Claims the firm corrects before filing because Ledger Room held them, with the fine each would have drawn.

Real numbers we get

- Form 401 already counts each kind of input voucher. Each period the firm exports those counts for all its clients.
- That gives the firm's own number of paper invoices a month for the keying figures in appendix A7.
- The agent's first run on real held vouchers, with the firm's own earlier vouchers to look up.

Form 401 (VAT return) reports the number of input vouchers by kind; one public fee table already bills by that count.

APPENDIX A10 · QUESTIONS WE EXPECT

Eight questions a judge would ask, answered

Question	Answer
What does the agent add, if code checks everything anyway?	A second reading. On the same 30 held vouchers, a code-only shortcut cleared 9 and let 2 wrong ones through; with the agent's readings code cleared 10 and let none through. 9 of the 10 passed on the sum check; the agent never decides alone (appendix A2).
Why a vision model, if OCR scores the same?	Because on real paper it reads far better. Real box: OCR alone got 60% of fields wrong, the 9B alone 27%, both checked against each other in code 21.5%. On clean synthetic print the two score the same.
Why not send the photos to a cloud model?	Because client records stay in the office. Any cloud reader, even in Taiwan, is outsourcing the firm must supervise (Art. 10); a Gemini reader also sends data abroad (Art. 11). The box: 0 outside connections on the B70.
What if the model misreads?	It never decides. On the real company's 113 invoices the 9B misread 10 and all 10 went to a person (0 through, on the three fields its old ledger keeps). On 58 public photos with strict reading: 0 through.
How does it do on layouts it has never seen?	It holds what it is unsure of. 53 synthetic documents in six new layouts, first run: 11 of the 13 with a misread went to a person; rule F1 0.84, 1.00 after the reader fix (appendix A5). The agent then cleared 6 of the 12 it held, none wrong.
Why this card?	The 9B reader and the 27B agent stay loaded together on one card: 21.8 GB live · 23.0 GB peak (agent run, strict reading), of the 30.3 GB the driver reports. Same code, same machine: the 9B takes 3.6 s a voucher on the card and 9.8 s on the host's CPUs (earlier run on GPU.0: 3.8 s); one card read about 1,000 documents an hour for 70 minutes with no restarts.
Is the box you price the box you measured?	Same GPU. We measured Intel's own Arc Pro B70 boards (230 W) on the organizers' Xeon 6787P host; the box we price uses ASUS's UGen B70, the same GPU on a 310 W board, with a Core Ultra 5. We measure the ASUS-built box next.
How were the tests kept fair?	Keys were frozen before runs and first runs are the official numbers. Real data: 336 vouchers of one company, 58 public photos, 193 foreign receipts. Next: a pilot firm.

The likely questions, answered here.

APPENDIX A11 · SCOPE

Scope today, and what we measure next

Real data

- Measured: one real company's box, 336 vouchers, keyed by 4 labelers from the page images, frozen before any read. Next: score against the return a pilot firm files (appendix A9).
- Measured: 58 public Taiwan photos with strict reading, 0 let through; 17 keyed before any read and run once. Next: the firm's own scans and phone photos.
- Measured: strict reading, two agreeing printed sources for every filed field. Next: the same rule on every voucher of the pilot.

Test sets and the B70

- Measured: synthetic sets with planted issues, one per rule, keys frozen first. Next: the issues a real firm actually has.
- Measured: six new layouts (blind3), rule F1 0.84 on the first run, 1.00 after the reader fix. Next: the firm's own layouts.
- Measured: Intel's own Arc Pro B70 boards on the organizers' Xeon host, the same GPU as the ASUS UGen B70. Next: the ASUS-built box we price.
- Measured: 70 minutes on two cards, 0 restarts after the zoom-strip fix. Next: a one-image reproduction for an OpenVINO report.

Product

- Measured: the agent on 52 held synthetic vouchers, strict reading, 19 cleared, 0 wrong. Next: the firm's real held vouchers, with its earlier vouchers to look up.
- Measured: the 401 filing pair passes the MOF offline checking program (sample client). Next: the 403 pair and the firm's own platform download.
- Measured: keying time saved on one company's paper invoices (appendix A7). Next: the firm's own keying hours and form 401 counts (appendix A9).
- Next: seconds a person spends on each held voucher, timed in the pilot.

Reproduce: pytest -q · python -m eval.bench_b70 · python -m eval.bench_concurrency · python -m eval.offline_check · python -m eval.strict_rescore · Source code on GitHub · code MIT, model weights Apache-2.0

APPENDIX A12 · WHERE EACH MAIN-SLIDE NUMBER COMES FROM

Sources for slides 2 to 11

Slide	Numbers	File or page
2, 5	15 of 113 rows wrong; 10 misread, 0 through, 53 to a person, 60 without a person and none wrong	eval/results_real_company.json → old_ledger_vs_ledger_room
2	0.5×, 1×, 3× the tax; 85,288 cases, 25,600 fined, NT$3.5 billion; no fine before any inquiry	MOF penalty reference table (law-out.mof.gov.tw GL009978); MOF Fiscal Statistics Yearbook 2025, Table 3-29 (service.mof.gov.tw); Tax Collection Act Art. 48-1
5, A4	58 public photos, 0 through; MOF checker PASS (401 pair, sample client); 2,383 reads in 70 minutes, four test sets, 0 restarts	eval/results_real_tw_strict*.json, results_real_tw2_strict*.json; MOF offline VAT checking program 11500.01 (run 2026-11-08); eval/results_longrun_zoompad.json
A5	new layouts 11 of 13 held, 40 of 326 fields; rule F1 0.84, 1.00 after the fix	eval/blind3_first_run_per_doc.json; eval/results_tiered.json → sets.blind3; eval/results_after_fix_r9_blind3.json
6, A2	30 held → 10 cleared, 0 wrong; code-only 9 cleared, 2 wrong; blind3 12 → 6; 8 s	Agent run on the organizers' host, default reading: eval/results_agent_b70.json; strict re-run: eval/results_agent_b70_strict.json; answer keys eval/expected_*.json
7	0 outside connections	eval/results_offline_b70.json; law-out.mof.gov.tw GL011130
8, A6	21.8 GB live · 23.0 GB peak (agent run, strict reading); 3.6 s and 9.8 s a voucher (earlier GPU.0 run: 3.8 s); about 1,000 an hour a card; two cards 2,042 an hour (1,031 + 1,011), 91% busy; 194 W, 187 Wh per 1,000; p50 3.4 s, p95 4.1 s (70 min: 3.5 / 4.5 s)	eval/results_one_card.json; eval/results_agent_b70_strict.json → memory_gb.peak; eval/results_same_host_cpu_vs_b70_live.json; eval/results_concurrency.json → rows.0 (power, latency_s); eval/results_longrun_zoompad.json → rows.0; eval/results_longrun_zoompad_util.json → cards.*.mean_busy
9, A7	85,288 cases; 8 catches a year pay box + licence (1 return in 38), 2 the licence (example); 0.5×, 1×, up to 5×; NT$20 a client, NT$12,000; box NT$91,501–99,245; 53 of 113 held; sample period 18 of 18 caught, 0 false holds; A7: 24% (9 of 38), 18.4–28.5 months; firms	eval/results_real_company.json; eval/results_period_check.json; coolpc.com.tw, asus.com/tw (2026-11-08); €1,799 (guru3d); help.dext.com; MOL 2025 wage survey; CTEE 2025-11-26; FSC 2025; MOF penalty table; VAT Act Art. 51; Economic Daily 2026-06-11
10, A8	0 of 4; 1.715 million SMBs targeted; Armor EdgeAI, IFA; ASUS parts NT$21–24 million (320 firms), NT$0.73–0.82 billion (11,035)	experthub.asus.com (read 2026-11-07, 11-08); news.cnyes.com 6537135; bnext.com.tw/article/91534; appendix A7 (box parts, firm counts)
11	Proof points and next steps: 0 of 113; 19 of 52, 0 wrong; 23.0 GB; 3.6 / 9.8 s; about 1,000 an hour; 0 connections	As slides 5–8 (19 of 52: eval/results_agent_b70_strict.json); appendix A11, appendix A9

numbers_table.md (next to build_deck.py) lists every number on every slide with its file and field; the build asserts the headline numbers against these files.